

1. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$\frac{54.00 + 33.00i}{-1.00 + 8.00i}$$

- A. $a \in [2.5, 3.5]$ and $b \in [-8, -6]$
 - B. $a \in [-54.5, -53.5]$ and $b \in [3, 4.5]$
 - C. $a \in [-5.5, -4]$ and $b \in [5.5, 6.5]$
 - D. $a \in [2.5, 3.5]$ and $b \in [-466, -464]$
 - E. $a \in [209.5, 210.5]$ and $b \in [-8, -6]$
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2. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$(5.00 - 2.00i)(-8.00 - 9.00i)$$

- A. $a \in [-59, -56]$ and $b \in [-31, -26]$
 - B. $a \in [-23, -18]$ and $b \in [-63, -59]$
 - C. $a \in [-23, -18]$ and $b \in [57, 62]$
 - D. $a \in [-59, -56]$ and $b \in [21, 33]$
 - E. $a \in [-45, -37]$ and $b \in [15, 19]$
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3. Simplify the expression below and choose the interval the simplification is contained within.

$$8 - 11 \div 20 * 14 - (7 * 17)$$

- A. $[-120.1, -118.3]$
- B. $[-111.9, -109.5]$
- C. $[125, 127.2]$
- D. $[-114.8, -111.8]$

E. None of the above

4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$\sqrt{\frac{53361}{441}}$$

- A. Whole
 - B. Integer
 - C. Rational
 - D. Irrational
 - E. Not a Real Number
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5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\sqrt{\frac{-1989}{0}}i + \sqrt{234}i$$

- A. Rational
 - B. Irrational
 - C. Nonreal Complex
 - D. Pure Imaginary
 - E. Not a Complex Number
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